testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Speed Time & Distance

Speed, Time & Distance

Basic Formulas:

- Speed = Distance / Time
- Time = Distance / Speed
- Distance = Speed × Time

Conversion of Units

1. 1 h = 60 min = 60 × 60 = 3600 sec.
2. 1 km = 1000 m.
3. 1 mile = 1.606 km or 1 km = 0.6214 mile; 5 mile = 8 km.
4. 1 yard = 3 ft.
5. p km/hr = $p \times (\frac{1000}{3600})\ m/sec. = p \times (\frac{5}{18})\ m/sec.$
6. p m/sec = $p \times (\frac{60 \times 60}{1000})\ km/hr. = p \times (\frac{18}{5})\ km/hr.$

Special Cases:

- If Distance is Constant: Speed is inversely proportional to Time

$$\text{Speed} \propto \frac{1}{Time} \Rightarrow S_1 \times T_1 = S_2 \times T_2$$

- If Time is Constant: Distance is directly proportional to Speed

$$\text{Distance} \propto \text{Speed} \Rightarrow D_1 / D_2 = S_1 / S_2$$

- If Speed is Constant: Distance is directly proportional to Time

$$\text{Distance} \propto \text{Time} \Rightarrow D_1 / D_2 = T_1 / T_2$$

Average Speed:

- Average Speed = $\frac{Total\ Distance}{Total\ Time}$
- Average Speed = $\frac{D_1 + D_2 + D_3 + ...}{T_1 + T_2 + T_3 + ...}$

Cases for Average Speed :

Case 1: When distance is constant

- Average Speed = $\frac{D + D}{T_1 + T_2} = \frac{2D}{T_1 + T_2}$
- Average Speed = $\frac{2S_1S_2}{S_1 + S_2}$

For 3 equal distance travel:

- **Average Speed =** $\frac{3S_1S_2S_3}{S_1S_2 + S_2S_3 + S_3S_1}$

Case 2: When time is constant

When two D_1 and D_2 are covered at speeds S_1 and S_2, respectively the average speed V_{avg} can be calculated as:

- **Average Speed =** $\frac{S_1 + S_2}{2}$

Relative Speed:

- Same direction: Relative Speed = $|S_1 - S_2|$
- Opposite direction: Relative Speed = $S_1 + S_2$

Problems on Trains:

- **When crossing a stationary object:**
  - Time = Length of Train / Speed of Train
- **When crossing a moving object in the same direction:**
  - Time = Length of Train / (Speed of Train – Speed of Object)
- **When crossing a moving object in the opposite direction:**
  - Time = Length of Train / (Speed of Train + Speed of Object)
- **When crossing a platform/bridge:**
  - Time = (Length of Train + Length of Platform) / Speed of Train
- **When two trains cross each other:**
  - Same direction: Time = $(L_1 + L_2) / (S_1 - S_2)$
  - Opposite direction: Time = $(L_1 + L_2) / (S_1 + S_2)$

Boats and Streams:

- Downstream Speed = Speed in still water + Speed of current
- Upstream Speed = Speed in still water – Speed of current
- Speed in still water = (Downstream Speed + Upstream Speed) / 2
- Speed of stream = (Downstream Speed – Upstream Speed) / 2
- Time for Downstream Journey = Distance / (x + y)
- Time for Upstream Journey = Distance / (x – y)

Some Important cases:

Case 1: If the total time taken by the boat to row a distance of D and reach back to its initial position is T, then the distance between the two places is:

- Distance = $\frac{T(B^2 - S^2)}{2B}$

Where T is the total time taken, B is the speed of the boat in still water, and S is the speed of the stream.

Case 2: If it takes T hours more to go to a point upstream than downstream for the same distance, then:

- Distance = $\frac{T(B^2 - S^2)}{2S}$

Case 3: If a boat travels a distance downstream in T_1 hours and returns the same distance upstream in T_2 hours and S is the speed of the stream, then:

- Speed of the boat in still water = $\frac{S(T_1 + T_2)}{T_1 - T_2}$

Case 4: If the time taken by the boat to row the same distance downstream is T_1 and upstream is T_2. Then,

The ratio of the speed of the boat to the speed of the stream = $\frac{B}{S} = \frac{T_1 + T_2}{T_1 - T_2}$

Total Time for Downstream + Upstream Journey: If distance downstream = d_1, upstream = d_2:

- $t = (d_1 / (x + y)) + (d_2 / (x - y))$

Average Speed (round trip): If the distance is the same both ways:

- Average Speed = (2xy) / (x + y)

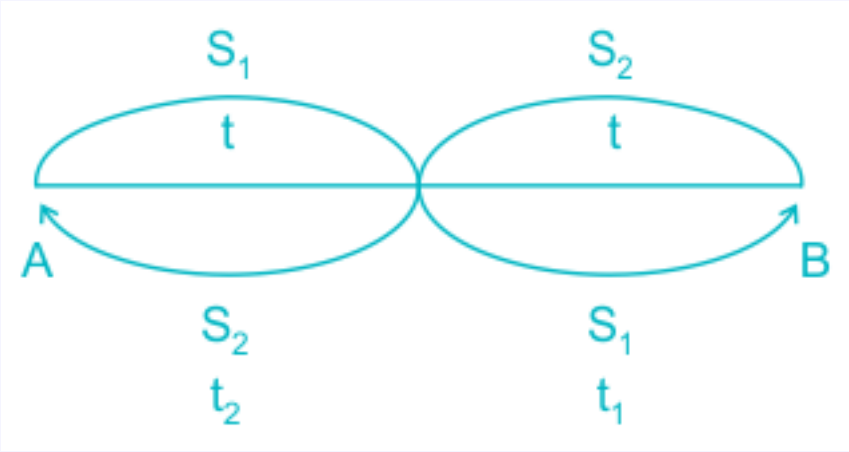
Race and Circular Tracks:

- Time to meet (same direction) = Track Length / (Faster Speed – Slower Speed)

- Time to meet (opposite direction) = Track Length / (Sum of Speeds)
- Time to meet again at the starting point = LCM(Time taken by each to complete one round)
- Number of distinct meeting points:

  N = (Time to meet at starting point) / (Time to meet anywhere)

Two Men Meeting at a Point And Cross Each Other



$$1.\text{Total Time (T)}: \quad T = \sqrt{T_1 \times T_2}$$

$$2.\text{Ratio of Speeds (S1/S2)}: \quad \frac{S_1}{S_2} = \sqrt{\frac{T_2}{T_1}}$$

$$3.\text{Distance (D)}: \quad D = S_1T_1 + S_2T_2$$

Here,

S1 → Speed of man starting from A
S2 → Speed of man starting from B
T → Time after which they meet each other.
T1 → Time taken by man 1 to reach its destination after crossing each other.
T2 → Time taken by man 2 to reach its destination after crossing each other.
D → Total distance from A to B.